

DATE: September 20, 2026

SUBJECT: Analysis of 2-Year Public Enrollment Trends and Federal Grant Allocation

Executive Summary

- Enrollment: First-time, full-time undergraduate enrollment in the defined public two-year subgroup changed by -16.65% between academic years 2010–2011 and 2015–2016 within the retained panel.
- Grant allocation: The observed and simulated cross-state 90/10 ratios of aid per enrolled student are 1.63 and 1.12, respectively. Holding enrollment fixed, simulated minus observed grant spending is \$134.3 million in the retained sample for 2015–16.

Data & Methodology

The analysis uses a complete-case balanced panel containing 5,148 undergraduate institutions and 30,888 institution-year observations from unrevised IPEDS files. The study covers academic years 2010–2011 through 2015–2016 and restricts geography to the 50 states; Washington, DC, and the other excluded areas are outside the study scope. Year labels in the enrollment figure denote the academic year’s starting year.

For enrollment (`SCUGFFN`) and federal grants (`FGRNT_T`), values with corresponding item-status codes J, L, N, or P are treated as missing. This is not an R-only restriction on all fields. Institutions are retained only if all required variables are observed and the geographic, undergraduate, and classification restrictions are satisfied in every year. The results therefore describe the retained institutions rather than the full population. The analysis does not establish whether this selection overstates or understates the population-wide enrollment change.

The assignment’s “public two-year” category is operationalized as public undergraduate institutions with `CONTROL` equal to 1 and `ICLEVEL` in $\{2, 3\}$. Institutional level is used as a proxy for the assignment’s non-bachelor’s-granting definition; the subgroup is not limited to institutions coded `ICLEVEL` equal to 2. Subgroup membership is selected separately in each year, so the balanced institution panel does not necessarily imply an unchanged set of public two-year institutions.

2-Year Public College Enrollment Trends

Total first-time, full-time undergraduate enrollment in the defined public two-year subgroup was 806,760 in 2010–2011 and 672,435 in 2015–2016, a change of -16.65% (Figure 1). These totals measure the specified entering cohort, not all enrolled students. The comparison is descriptive: it does not identify the causes of the change or establish a long-term structural trend.

Current Federal Grant Distribution (2015–16)

The grant analysis includes all retained undergraduate institutions in 2015–16, not only the public two-year subgroup. Each state rate is total federal grant aid for the measured first-time, full-time undergraduate cohort divided by total enrollment in that cohort at retained institutions in the state. The denominator includes students not awarded federal grants; this is not the average grant among recipients. States identify institutional locations, not students’ states of residence. Dollar amounts are not adjusted for inflation.

Aid per enrolled student was \$2,315.17 in New York and \$1,364.22 in Vermont (Figure 2). Across the 50 included states, the unweighted mean of state rates was \$1,880.22, the sample standard deviation was \$373.97, and the 90/10 ratio was 1.63 (Figure 3). States receive equal weight in these statistics; the mean is not the pooled, enrollment-weighted average. The 90/10 ratio is the 90th percentile divided by the 10th percentile of state rates, with linear percentile interpolation. Standard deviations use the sample formula. These differences do not, by themselves, establish geographic favoritism or inequity; the comparisons do not adjust for student need, eligibility, or institutional composition.

Simulated Policy Impact

The simulation applies the specified allocation formula at the institution level:

$$G_i^{\text{sim}} = 1750N_i + 0.15N_i^2,$$

where N_i is first-time, full-time undergraduate enrollment at institution i and G_i^{sim} is its simulated federal grant amount in dollars. Observed 2015–16 enrollment and the retained institutional sample are held fixed. The allocation is not rescaled to preserve the observed total budget. The exercise does not model changes in enrollment, institutional behavior, or implementation expenses.

The observed and simulated standard deviations of state aid rates are \$373.97 and \$120.00, respectively. The corresponding highest-minus-lowest ranges are \$1,629.15 and \$711.37, and the 90/10 ratios are 1.63 and 1.12 (Figures 4 and 5). These are comparisons of cross-state dispersion, not a welfare or equity assessment.

Simulated minus observed grant spending is \$134,258,227.65 (\$134.3 million) within the retained sample for 2015–16. Positive values indicate additional grant spending; negative values indicate lower grant spending. This is a single-year, sample-specific difference, not a cumulative amount or a nationwide implementation-cost estimate. Figure 6 identifies states with increases and decreases in grant totals at their retained institutions, rather than states that are better or worse off overall.

Conclusion

The analysis documents enrollment changes within the retained panel and compares observed grant allocations with a formula-based, fixed-enrollment simulation. The findings describe cross-state dispersion and simulated grant spending within that sample. They do not establish the causes of enrollment changes, the direction of selection bias, or the welfare effects of the alternative allocation.

Appendix: Figures

Notes applying to all figures: Results describe retained institutions only. Enrollment refers to first-time, full-time undergraduates; public two-year status follows the operational definition in the methods. In Figures 2–5, per-student rates divide cohort grant totals by cohort enrollment, including nonrecipients, at all retained institutions in each state. Cross-state comparisons give each state equal weight. Simulated amounts use fixed observed enrollment and are not constrained to preserve observed grant spending. Figure 6 classifies changes in grant totals, not welfare effects. Monetary amounts are in dollars without inflation adjustment.

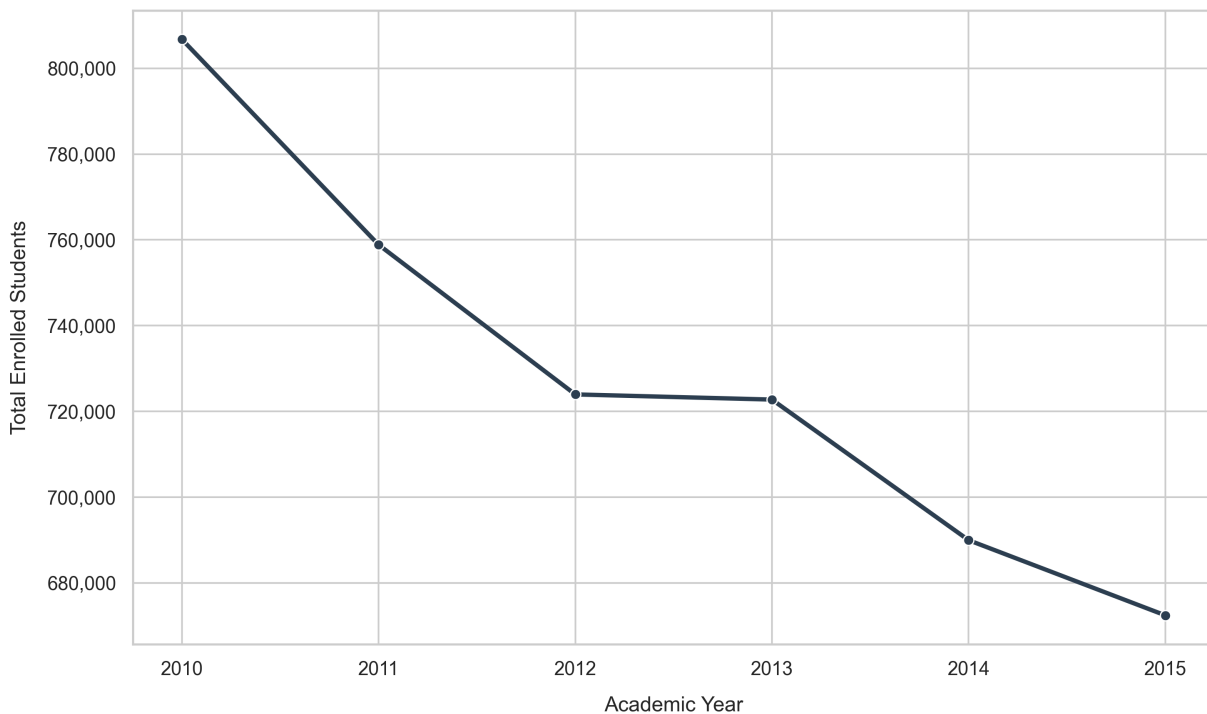


Figure 1: First-Time, Full-Time Undergraduate Enrollment in the Defined Public Two-Year Subgroup (2010–2015; Academic-Year Starting Years)

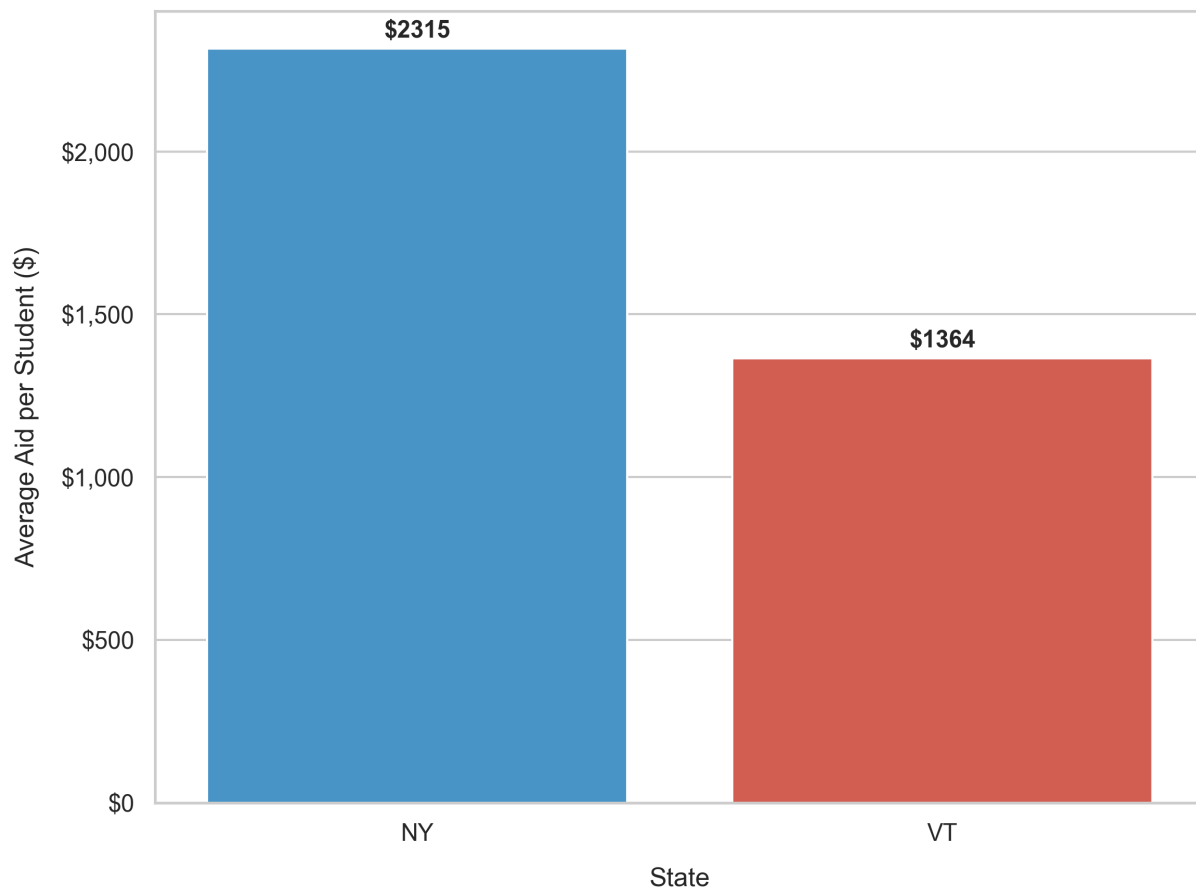


Figure 2: Federal Grant Aid per Enrolled Student at Retained Institutions: New York and Vermont (2015–16)

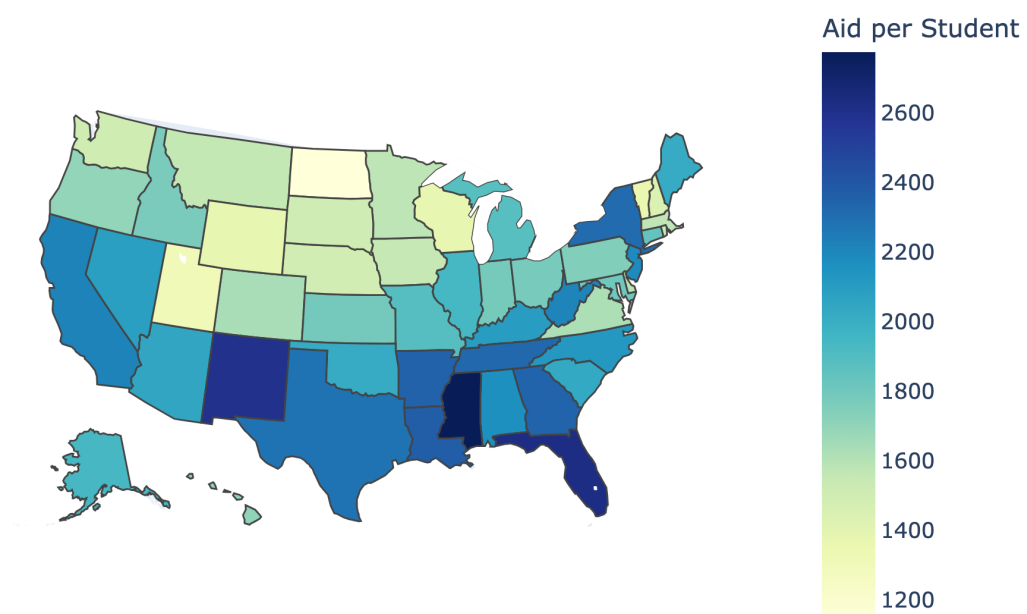


Figure 3: Federal Grant Aid per Enrolled Student at Retained Institutions, by State (2015–16)



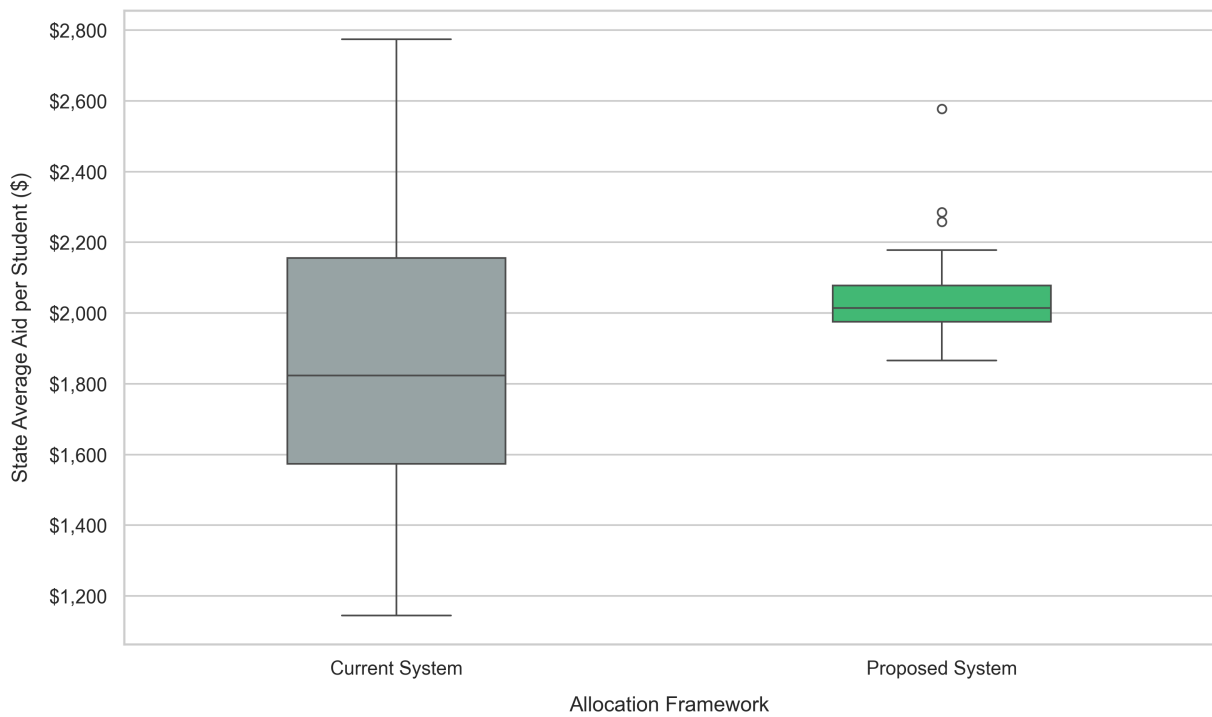


Figure 5: Distribution of State-Level Aid per Enrolled Student: Observed and Simulated (2015–16)

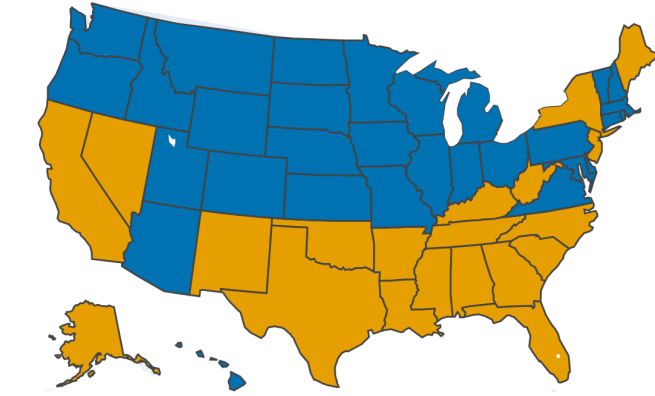


Figure 6: States with Increases and Decreases in Simulated Grant Totals at Retained Institutions (2015–16)